NUCLEAR FUEL CYCLE

Uranium from Rare Earth Deposits

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- A large amount of uranium is in rare earth deposits, and may be extracted as a by-product.
- Higher uranium prices and geopolitical developments would enhance the economic potential for recovering these.
- Rare earths are essential for many modern technologies. Growth in electric car use will depend, in part, on the availability of rare earths.

In addition to the 7.9 million tonnes of uranium in the world's known recoverable resources, there are substantial amounts comprising what is known as 'unconventional resources'. Such unconventional resources, from which uranium can be produced in conjunction with other metals, have accounted for over 11% of historical uranium production. Rare earth element (REE) deposits are one such unconventional resource. REEs have unique catalytic, metallurgical, nuclear, electrical, magnetic and luminescent properties, and play a critical role in the application of many modern technologies, including magnetic resonance imaging (MRI) machines, satellites, batteries, LED screens and solar panels. China is the leading supplier of REEs, giving rise to commercial pressure for development of deposits elsewhere.

In the USA, and likely elsewhere, some uranium producers are recovering small amounts of rare earths as by-product of uranium production. This can make a significant difference to economics when uranium prices are low.

Privacy - Terms

REEs are a set of 17 chemical elements in the periodic table, specifically the 15 contiguous lanthanides plus the lighter scandium and yttrium. Scandium and yttrium are considered REEs since they tend to occur in the same ore deposits as the lanthanides and exhibit similar physical and chemical properties. REEs are in fact relatively abundant in the Earth's crust, but are rarely found in concentrations that are economically exploitable. Different REEs have similar chemical compositions, meaning that they readily bond with one another, resulting in a complex extraction process. REE resources occur in four primary geological settings: carbonatites; ion-absorption clay deposits; igneous systems; and monzanite-xenotime placer deposits. REE resources are usually reported as rare earth oxides (REO).

Types of REE

Economically and geologically there is a distinction between **light REEs**, or cerium earths, and **heavy REEs**, or yttrium earths.

Light REEs

- Scandium, lanthanum, cerium, praseodymium, neodymium, promethium, samarium, europium, gadolinium – elements 21, 57-64.
- Primarily produced from carbonatites and placer deposits.
- Large orebodies known and being exploited around the world.

Heavy REEs

- Yttrium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, lutetium – elements 39, 65-71.
- Primarily produced from ion-absorption clays.
- Resources of heavy REEs are typically rarer, smaller and less concentrated.
- Most of the current supply of heavy REEs originates in the 'ion adsorption clay' ores of southern China. Some of these provide concentrates containing about 65% yttrium oxide, with the heavy lanthanides. Uranium ores from Ontario have

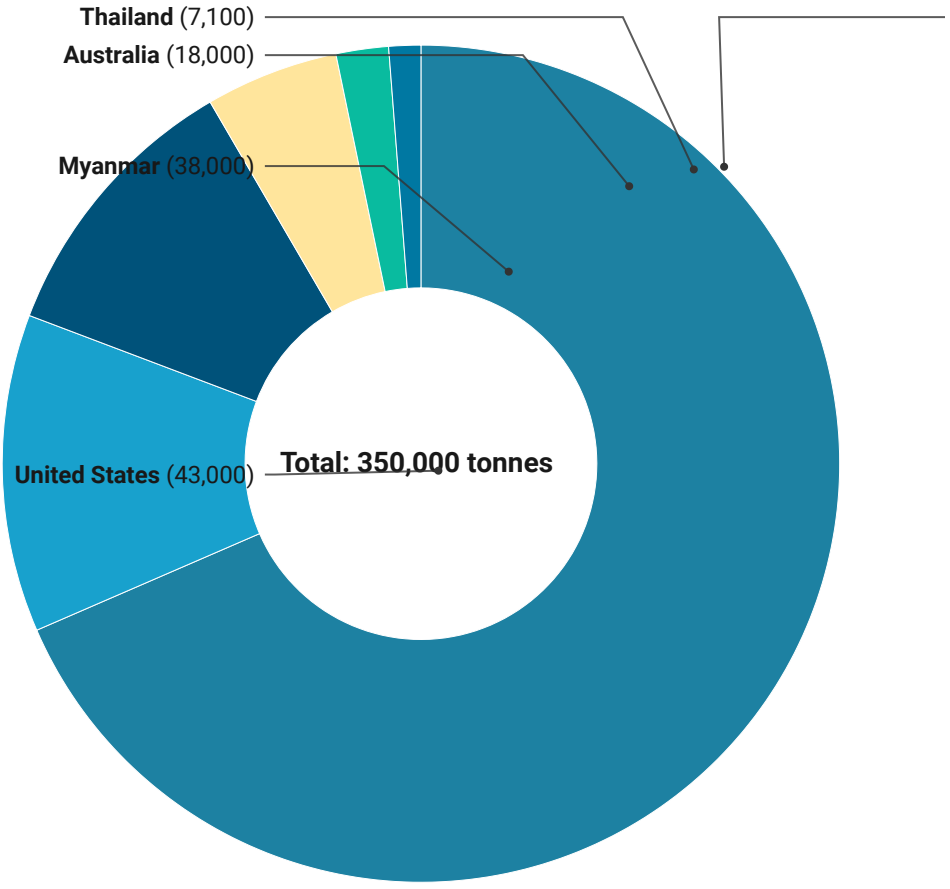
occasionally yielded yttrium as a by-product. Xenotime incorporates yttrium and heavy REEs.

Samarium, europium and gadolinium can fall into either category, or with terbium and dysprosium form a **middle REEs** category. In 2015 Greenland Minerals said that neodymium, dysprosium, terbium, europium and yttrium were critically important for clean energy uses and had high supply risk.

Production & reserves: China's dominance

In 2023 China produced about 70% of the world's 350,000 t of REEs, mostly from Inner Mongolia.

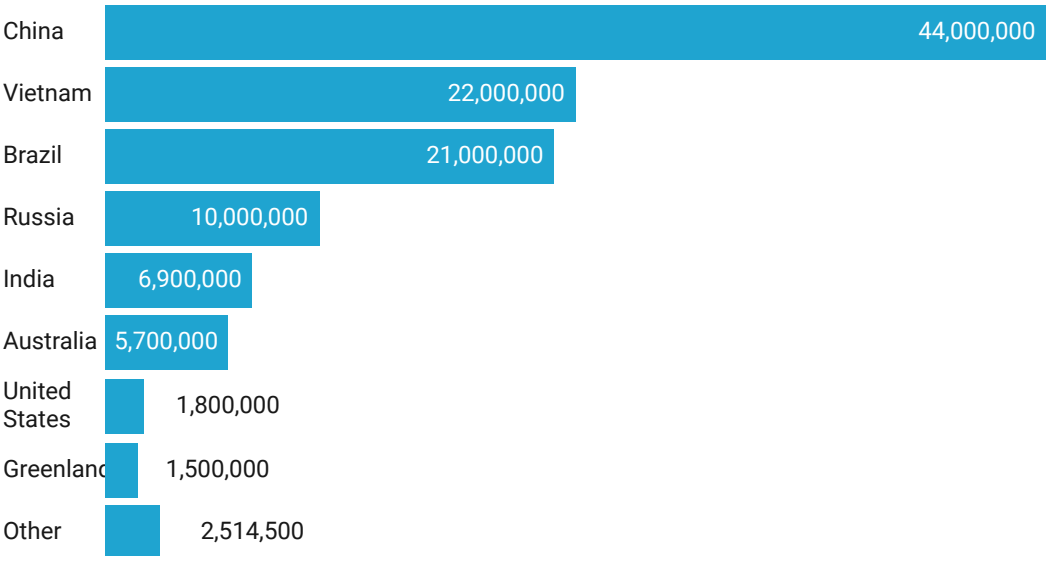
Rare earth element production by country 2023



Source: United States Geological Survey (USGS)

Considering REEs alone, China has 44 million tonnes (40%) of the world's 110 million tonnes of REO reserves, followed by Vietnam with 22 million tonnes (20%) and Brazil with 21 million tonnes (19%).

Rare earth element resources by country 2023



Source: USGS

All of the world's heavy REEs (such as dysprosium) come from China. Most REE is by-product from Chinese sources such as the polymetallic (iron-REE-niobium) Bayan Obo deposit in Baotou city. There are also limited surficial clay deposits in southern China, and two in Jianxi province are exploited: Longnan, with heavy REE similar to xenotime, and Xunwu, with light REE particularly lanthanum but with little cerium.

China cut back exports of all REE from the 35,000 t level in 2010, and banned exports of terbium and dysprosium. This led to recommissioning of Molycorp's Mountain Pass mine in California, and enhanced interest in bringing new orebodies into production, including Lynas' Mount Weld in Western Australia (now about 11,000 t/yr), Nolans Bore in Central Australia and Kvanefjeld in Greenland, the latter two with significant uranium content.

In April 2011 China's Ministry of Industry & Information Technology (MIIT) under the National Development & Reform Commission (NDRC) set a production quota for REO of 93,800 tonnes, up 5% on 2010, with a quota on REO smelting of 90,400 tonnes. No new mining licences were to be granted before mid-2012. MIIT said that no extra production beyond the quotas would be allowed. China's export quotas and tariffs on REE ended in 2015, and in 2017 the production quota was set at 105,000 tonnes.

From 2011 Inner Mongolia Baotou Steel Rare-Earth (Group) acquired five rare-earth separation companies licensed by the Inner Mongolia government. It is now known as Baogang, based in Hexi Industrial Zone, Baotou, Inner Mongolia. Its main asset is the

Baiyunebo mine. This was part of the national government's effort to consolidate China's REE sector, gain more influence over pricing in the global market and ensure sustainable growth in the industry.

Production methods

In China the main REE mine production is from carbonatite-related deposits such as bastnasite, ionic clays, and monazite/xenotime, the last having more uranium. Outside China, monazite, xenotime, apatite are the main sources. In Greenland, the complex steenstrupine is the major REE mineral and has around 0.4% uranium. Bastnasite is a carbonate-fluoride mineral, monazite and xenotime are phosphates. All contain more than 50% REEs and traces of uranium or thorium which may be economically recovered as by-product. (Apart from Elliot Lake to 1990s there has been no commercial recovery of REE from uranium mining.) In USA monazite has about 14% heavy REEs, compared with one-tenth of that in bastnasite.

Typically a REE mine will produce a REE carbonate which then needs to be treated for REE separation, followed by refining of the REEs. There are three main processes: acid bake, caustic cracking, acid leach. Acid bake is used by Lynas and Arafura and can readily be adapted for uranium recovery. Caustic cracking is used for pure monazite and is not ideal for uranium recovery. The acid leach flowsheet will be used by Greenland Minerals to recover uranium which is necessary for a saleable REE product. A significant problem with higher levels of uranium in the REE carbonate is actinium-227, from all processes. It is a uranium-235 decay product but is chemically very similar to lanthanum, so reports with it.

Particular deposits and projects

Kvanefjeld, Ilimaussac complex, southern Greenland

Kvanefjeld is the main REE deposit with major potential for uranium production, with Sorensen, Zone 3 and Steenstrupfjeld orebodies in the same Ilimaussac intrusive complex. It was investigated intensively over 1955-86, then dropped for commercial reasons. Greenland Minerals & Energy, now Greenland Minerals, acquired the project in 2007 then with a JORC-compliant resource of 43,000 tU at 0.022% U with 6.5 million tonnes of REO at 1.07% in

Lujavrite. About 3.6% of REOs are terbium, dysprosium and yttrium, *i.e.* 'heavy'.

Resources

228,000 tU (May 2015, JORC compliant) in four deposits: Kvanefjeld, Sorensen, Zone 3 and Steenstrupfjeld. This includes 36,600 U measured resources at 0.0257 %U, 66,000 tU indicated resources at 0.0215 %U (all in the Kvanefjeld deposit), and 125,000 tU inferred resources (in the first three deposits), all with 0.015% U_3O_8 cut-off

11.14 million tonnes of REO including 0.40 million tonnes of heavy REO. Also 2.42 Mt zinc as sphalerite.

Relative to other world REO deposits (especially China), Kvanefjeld has a high proportion of terbium through to lutetium (elements 65-71). There is considerable further mineral potential in the immediate area – including up to 600,000 tU according to IAEA estimates.

The maiden ore reserves (JORC compliant, May 2015) are 43 Mt proven with 352 ppm U_3O_8 (12,834 tU) and 500 ppm heavy REO, and 64 Mt probable with 368 ppm U_3O_8 (19,970 tU) and 490 ppm heavy REO. These are in the upper part of the Kvanefjeld deposit. The 2018 'Red Book' lists Greenland's reasonably assured uranium resources recoverable as 66,800 tU, and in situ as 102,800 tU in the <$260/kgU category, and total identified resources as 148,200 tU recoverable and 227,700 tU in situ in the same cost category, all attributable to Kvanefjeld.

Process

The mine and concentrator with flotation circuit would produce a REE-U concentrate containing 20-25% REO plus zinc concentrate and fluorspar by-products. Then the refinery would employ atmospheric sulfuric acid leach to produce uranium oxide by-product, then stronger acid leach followed by solvent extraction recovery, and precipitation to separate REEs. Primarily it would produce a critical rare earth concentrate (Nd, Pr, Eu, Dy, Tb, Y), with lanthanum and cerium by-products. Production is envisaged as 7900 tonnes of critical mixed REO per year, 434 tU, 15,100 t lanthanum and cerium oxides, 15,000 t zinc concentrate and 16,000 t fluorspar per year.

Metallurgical flowsheets for both concentrator and refinery were developed in 2013, and have been modified since. In collaboration with Shenghe, test work over 2017-18 at China's Institute for Multipurpose Utilisation of Mineral Resources (IMUMR) at Chengdu and Baotou Meng Rong Fine Materials Co Ltd (BTMR) achieved

much higher concentrate grades than in the original feasibility study. This allows consideration of selling the concentrates, rather than further processing. In January 2019 BTMR was selected to continue the development and validation of its flotation circuit.

The concentrator is planned to be at the head of Narsaq Valley adjacent to the mine, and the refinery about one kilometre down the valley. A new port will be at the bottom of the Narsaq Valley.

Start-up costs for a 3 Mt per year plant were initially estimated at $1.121 billion for mine, concentrator and refinery, plus $240 million for infrastructure in the May 2015 feasibility study. This estimate was reduced in 2016 and then the study was further reworked in collaboration with Shenghe Resources so that in 2019 the company announced that capital costs including refinery were then estimated at $505 million, and rare earth production increased by 8% to 32,000 t/yr REO and about 400 tU/yr, giving a 37-year mine life considering only proven and probable reserves.

Chinese partners

The company had been seeking partners for the refinery, then envisaged offshore, and in March 2014 signed an agreement with NFC, a subsidiary of China Non-Ferrous Metal Mining Group (CNMC) to provide concentrates of heavy REOs to NFC's new 7000 tpa rare earth separation facility in China, then under construction. A further agreement with NFC was signed in April 2015. NFC's participation in the rare earth industry comes through its subsidiary Guangdong Zhujiang Rare Earths Company, which was the first group to carry out full separation of 15 rare earth elements in China. Uranium is not included in the NFC agreement but it appears that this agreement has lapsed and been replaced by that with Shenghe Resources.

In September 2016 Shenghe Resources Holding Co Ltd, a Shanghai-based rare earths company, acquired a 12.5% stake in Greenland Minerals & Energy for AUD 4.625 million. The company welcomed this as a 'strategic investment'. It has an option to increase its share.

The company entered a non-binding agreement with Shenghe in August 2018 for the offtake of total output of rare earth elements from Kvanefjeld in either chemical or mineral concentrate. In January 2019 Shenghe formed a joint venture company – China Nuclear Hua Sheng Mining – with subsidiaries of China National Nuclear Corporation (CNNC) for the trading and processing of rare earth minerals with radionuclides. Hua Sheng provides a formal means for importing REE-U concentrates into China and will be

Shenghe's designated agent for that. It will enable finalization of a strategic development plan for the Kvanefjeld project.

Licensing

In December 2011, a year after allowing the company's feasibility studies to include uranium, the Greenland government amended the company's exploration licence to include uranium. The company could then apply for a mining licence including uranium, then with a view to first production in 2016, followed by a long mine life. In November 2012 the Greenland government voted unanimously to support the project, including uranium, and in October 2013 it repealed the long-standing policy banning uranium development.

In 2013 the government noted that it is Denmark's responsibility to ensure that international conventions, such as non-proliferation, are respected, since Greenland remains part of the kingdom of Denmark and its defence and foreign policies are still determined by Copenhagen. The Additional Protocol to Denmark's safeguards agreement with the IAEA, specifically for Greenland, entered into force in March 2013.

In January 2016 the governments of Denmark and Greenland reached agreements concerning the export control and security of uranium and other radioactive substances from Greenland and the definition of competencies in the raw materials sector. The Danish parliament passed legislation on safeguards and export controls in June 2016, assuming responsibility for the application of international safeguards. Corresponding legislation was passed by the Greenland parliament in May. This creates the legal framework for uranium exports from Greenland.

In 2018 the company lodged with the government updated environmental and social impact assessments (EIA, SIA) for the Kvanefjeld project, as well as the feasibility study and JORC-code compliant mineral resource estimate and ore reserve statement. The maritime safety study is pending. Following elections in April 2018 the ruling coalition reaffirmed its support for rare earth and associated uranium extraction and acknowledged that the time to move the licensing process to the next stage of public consultations is imminent. The company has applied to the government for a mining licence to proceed with the project.

In November 2021 Greenland's parliament passed legislation to ban the mining of mineral resources with a uranium concentration higher than an arbitrary 100 parts per million. The law was put forward by the Inuit Ataqatigiit party that came to power in April. It serves to reverse initiatives, policies and legislation adopted by

successive governments over the past decade. The Company is seeking clarity as to how the new legislation will effectively modify existing approvals or authorizations. Uranium, if recovered as by-product of rare earth production, would contribute about 5% of project revenues. The rare earth reserves include uranium at 360 ppm.

R&D

The company is involved with the EU-funded EURARE programme, designed to support the development of REO deposits in Europe. The Kvanefjeld project was selected by EURARE as a major one for demonstration of REO production, and a 26-tonne ore sample (about 1.3% REO) was processed over 100 hours at the beneficiation pilot plant in Finland by simple froth flotation to yield 2 tonnes of concentrate. The next test work stage was to produce 15% REO, then the pilot plant hydrometallurgy at Aachen in Germany to give a mixed carbonate of 60% REO. At Aachen a separation plant will sort Pr&Nd from La&Ce and from other materials.

In the later optimization with Shenghe Resources, a 22-25% REO flotation concentrate resulted, and its refining was simplified. Annual production was projected as about 1430 t praseodymium oxide, 4270 t neodymium oxide, 40 t terbium oxide, 270 t dysprosium oxide and 450 t uranium oxide.

Mt Weld, Western Australia

Lynas Corporation has the high-grade Mount Weld mine and mill in Western Australia, with its Advanced Materials Plant in Malaysia. It has been the only commercial REE producer outside China. Monazite is the primary rare earth mineral and no uranium is involved.

Mountain Pass, California

The Mountain Pass deposit was discovered by a uranium prospector in 1949. It dominated worldwide REE production from the 1960s to the 1990s, and was recommissioned to again be a major producer from 2014. Molycorp Inc spent $781 million on modernising and expanding its plant, which reached 19,000 t/yr capacity in 2012 – slightly more than US demand – and 20,000 t/yr in 2014. However, by June 2015 Molycorp had accumulated debt of

$1.7 billion and filed for bankruptcy. When China restricted its exports in 2010 Molycorp had a market value of $6 billion.

Remaining resources in carbonatite are about 1.6 million tonnes REE averaging 8.24%, with 5% cut-off. It produced both heavy and light rare earth concentrates. The company made no mention of uranium as by-product, but thorium is present in monazite.

Molycorp was restructured in 2016 as Neo Performance Materials. In 2017 the mine was sold for $20.5 million to a consortium – MP Mine Operations – backed by China's Leshan Shenghe Rare Earth Co, with 10% equity and with JHL Capital Group holding 65%. It is operated by MP Materials and ships concentrate for refining in China. It produces about 15% of world supply of rare earths. A second stage of recommissioning involves processing facilities which will make the operation self-sufficient from about 2022.

Eco Ridge, Elliot Lake, Ontario

Pele Mountain Resources had received mining leases for its Eco Ridge uranium and REE orebody in Canada, 11 km east of Elliot Lake. However, due to continuing weak uranium and rare earth prices, in June 2017 it sold the project to an unnamed buyer for $380,000.

In the decade to closure of the Stanleigh mine at Elliot Lake in 1996, the area produced REE as a commercial by-product of uranium production. The Eco Ridge deposit contains a full range of REE at 0.164% grade, including 9.5% heavy REE (including scandium, europium and gadolinium in this category, 7.4% if excluding them). Uranium grade is 0.041% U with cut-off grade 0.024%. Some 65% of the heavy REE report in the uranium leach solution without any additional milling cost. The deposit includes indicated resources of 36,600 t REO and 8690 tU, plus 56,900 t REO with 14,500 tU inferred resources (NI 43-101 compliant). An underground mine was envisaged.

A preliminary economic assessment in 2012 suggested uranium production of 10,500 tU over 11 years. However weakening REE prices coupled with the need for hardrock underground mining meant that the Eco Ridge project was deferred in favour of a proposed monazite processing facility at Elliott Lake, focused on neodymium and praseodymium. The monazite would come from mineral sands tailings in other countries. In March 2015 Pele announced an agreement with Sheng Kang Ning (Shanghai) Mining Investment Co. Ltd, a subsidiary of Shenghe Resources, to develop

monazite processing facilities to recover REE at Elliott Lake. However, this agreement lapsed when SKN could not get permission to use its proprietary separation technology outside China.

Teasdale Lake, Elliot Lake, Ontario

Appia Energy reports 3080 tU indicated resources and 7750 tU inferred resources in connection with rare earths in the Teasdale Lake zone of its Elliot Lake project.

Round Top, Texas

Texas Rare Earth Resources Corp (now Texas Mineral Resources Corporation) signed an agreement in April 2015 with Areva to take up to 300,000 lb U_3O_8 (116 tU) per year by-product uranium from its Round Top heavy rare earth project in Hudspeth County, Texas. Its 2013 preliminary economic assessment did not include the economic impact of uranium product and only a recommendation to perform a preliminary review of uranium potential, though 37,000 tU was identified in all classes of REE resources in rhyolite (with 70% being heavy REE). A July 2015 agreement established a joint venture with K-Technologies Inc to develop processing technology for REE.

In 2020 Texas Mineral Resources was planning to develop the deposit for heavy REE. It has a development agreement with USA Rare Earth LLC which will gain a 70% interest in Round Top project, possibly increasing to 80%. Planned production is 2212 t/yr total REO plus 8956 t/yr lithium carbonate.

Nolans Bore, Northern Territory, Australia

This is a deposit of light REE with 7500 tU content at 0.016% U with 1.2 Mt REO grading 2.6%, about 135 kilometres north of Alice Springs, NT. It also contains 5 Mt of 12.9% P_2O_5 and 0.27% thorium (Dec 2014). Perth-based Arafura Resources intends to develop it as a REE mine from 2019. The company now plans some processing about 10 km from the mine site followed by export through Darwin of an intermediate product to an offshore refinery for final processing to recover six rare earth products – lanthanum oxide, cerium oxide, NdPr oxide, SEG (Sm+Eu+Gd) oxide, HRE oxide ('heavy' rare earth oxide – Tb+Dy+Ho+Er+Tm+Yb+Lu+Y) and

cerium carbonate. Phosphate and uranium may be co-products. Thorium will be separated as an iron-thorium precipitate for long-term storage at the mine site as a possible future energy source.

Initial plans were to rail its concentrate at Whyalla, South Australia where it would be processed to produce 20,000 t/yr REO, 130 t/Uyr, plus phosphoric acid and gypsum co-products. The combined neodymium/praseodymium (NdPr) content of the ore means that NdPr oxide wolud generate about 80% of projected revenue, though these are both light REE. A definitive feasibility study was undertaken and in September 2016 the project was awarded major project facilitation status by the Australian government. In January 2021 the company announced completion of metallurgical test work and resulting process redesign.

Dubbo Zirconia Project, New South Wales, Australia

Alkane Resources' DZP has REE as potential by-product of zirconium production – 700,000 t zirconia as measured resources in 2011. These have 0.75% REO and 0.014% uranium – about 4200 tU, and the same in inferred resources. The 2012 reserve figures also have 0.75% REO at same tonnage but do not include uranium. With support from Export Finance Australia in 2020, Alkane subsidiary Australian Strategic Minerals (ASM) will seek strategic investors.

Zandkopsdrift, South Africa

Frontier Rare Earths in 2012 reported NI 43-101 resources of 950,000 t REE (80% as indicated resources) at this deposit in the Northern Cape Province, mostly in monazite. The resource includes 60-70 ppm U, which is to be removed for disposal onsite. A pre-feasibility study in mid-2015 reported 789,000 t reserves, of which 42% as proven.

Korean government-owned mining and natural resource investment company Korea Resources Corporation (KORES) acquired a 10% interest in the Zandkopsdrift project in December 2012 for a cash payment to Frontier of C$23.8 million ($24 million). KORES has an option to increase its stake to 50% on the condition that it arranges project finance for the entire development. Mining had been expected to commence in 2017.

Steenkampskraal, South Africa

During the 1950s and to 1963 Anglo American Corporation extracted about 50,000 tonnes of monazite concentrates with between 3.3% and 7.6% thorium. Great Western Minerals Group over 2011-2015 then invested to return the mine to production. Steenkampskraal Monazite Mine Ltd plans to operate the mine. In 2020, a NI 43-101 mineral resource estimate showed 87,000 t REO at high grade, including particularly neodymium, praseodymium and dysprosium. Earlier figures included 176,000 t monazite as indicated and 278,000 t inferred resources. Thorium was to be extracted from the mixed rare earth chloride concentrate, and stockpiled at a rate of about 360 t/yr. A private company, Steenkampskraal Thorium Ltd (STL), has the rights to this thorium and was planning to refine it, in collaboration with Thor Energy in Norway. (STL Nuclear is also designing a 100 MWt high-temperature pebble bed reactor – HTMR100 – to use the thorium.)

Cummins Range, Kimberly region, Western Australia

This is a deposit of light REE with inferred resource total REO of 85,000 t at 1.74%, uranium content of 0.0145% (600 tU), and 11.2% P_2O_5 at 1% cut-off. It is similar geologically to Mount Weld, but with greater uranium content. It has 127,000 t total REO inferred resources at 0.5% cut-off. Cummins Range was acquired in mid-2019 by RareX Ltd which then declared a maiden inferred resource of 13 million t at 1.13% total REO (147,000 t) containing a high neodymium and praseodymium ('NdPr') content of 22.1% and ultra-low thorium of 50 ppm ThO_2. The primary rare earth mineral is monazite.

Olympic Dam, South Australia

BHP Billiton's large Olympic Dam mining operation accesses a huge orebody, with the world's largest resources of uranium, though this is only a by-product of copper (along with gold and silver by-products). The orebody also contains some 53 million tonnes of REO (predominantly lanthanum and cerium), though these are sub-economic and there are no plans to extract them even as by-product at present. They remain in the tailings (as at many uranium mines).

Mary Kathleen, Queensland

This was a uranium mine in a REE deposit, and from which no rare earths were recovered commercially. It operated 1958-63 and 1975-82. The uranium grade was 0.10-0.15%, the REE grade about 4-5%, and the company initially had great difficulty keeping the REE assay in product below specified levels of 0.1%. It operated an acid bake pilot plant to recover REEs from the tailings stream, but this was not a commercial proposition and the REEs ended up in tailings. In 2015 the Queensland government was seeking expressions of interest in recovering them.

Dalur and Khiagda, Russian Federation

The Khiagda ISL uranium mine in Siberia is planning to recover REOs as by-product.

The Dalur ISL uranium mine in Trans-Ural region is planning to recover REOs and scandium as by-product.

White Mesa, USA

Energy Fuels Resources Corp (EFRC) is a uranium mining company in the USA that owns the White Mesa mill in Utah. The mill processes ore from mines in three states and already has a vanadium co-product circuit. EFRC is recovering REO concentrates as carbonate from some ores on a toll basis and says this will supply almost 10% of the US REE market. It plans to fully integrate REE carbonate production at the mill and is actively seeking supplies of monazite.

In March 2020, EFRC joined Canadian-based Neo Performance Materials to launch a US-European REE initiative to produce value-added REE products from natural monazite sands, a by-product of heavy mineral sands mined in southeastern USA by Chemours. In March 2021 EFRC commenced production of a mixed rare earth carbonate extracted from natural monazite sands at the White Mesa mill. A portion of this REE carbonate production is shipped to Neo's REE separations facility in Sillamae, Estonia (Silmet). Neo then processes the REE carbonate into separated REE materials for use in permanent magnets and other REE-based advanced materials. EFRC retains the uranium recovered from the monazite sands. It is aiming to source 15,000 tonnes of monazite per year,

which would utilize 2% of White Mesa's capacity and supply half of US demand for REE. US monazite contains about 55% REE, and about 22% of this is neodymium and praseodymium (light REE) and 8% heavy REE including dysprosium and terbium.

Notes & references

General Sources

Greenland Minerals
Pele Mountain Resources
Great Western Minerals Group
Arafura Resources Ltd
Texas Rare Earth Resources Corp
Alkane Resources
Geoscience Australia, Australian Mineral Atlas, Rare Earths
US Geological Survey, Mineral Commodity Summaries 2018

Related information

Supply of Uranium
Uranium Production by Country
Uranium from Phosphates